

Supplementary Extended Method Section and Lab Protocols

Study 1.

Method

Composition of the Voice Task

Measuring Voice Through Information Sharing. In the main task of the experiment, participant and (simulated by a computer, but previous to the task introduced to the participant as the confederate) Leader worked together on an information-sharing task which was originally designed as a group task by Stasser and Titus (1,2), and re-designed to fit our experimental setting and sample. The basic task is as follows: Participant and Confederate are assigned the Follower (participant) and the Leader (confederate) role. The team Leader has to decide on hiring a new Dean of the faculty. Team members both received separate information about three potential candidates for the job (A, B and C), as well as instructions to read the information carefully. It was explicitly stated that: as in real life, not all information had to be identical, and that there might be *different information presented to both team members*. The main task for the team Leader was to find the most suitable candidate for the job.

Both team members were asked to imagine they were part of the executive board of the university (the Leader in the role of CEO). It was explicitly explained that the team Leader would have to pick the best possible candidate using the information provided to the team. As predicted, the information was not identical for the Follower and the Leader: some information was available to both team members (shared), whereas some information was available only to one of them (unshared). Figure 1 shows the distribution of shared and unshared information across the Dean candidate profiles. This means that to arrive at the optimal choice for the team and faculty, they would have to share all their information with each other. It is important to point out that sharing information does not necessarily mean that

the correct information is processed and the best quality of a decision is reached. The main point of the classic Stasser and Titus' (1985) information sharing experiment, is that people generally focus on shared information, not on unshared information. In order to arrive at the optimal conclusion, Followers thus needed to 'think outside the box', and focus on novel information, not on what is already known to them. This, to us, contributes to finding an original and useful solution, and if Followers decide to voice that solution, they thus add value through originality, by voicing high-quality information.

Shared/Unshared Attributes			
Candidate -->	A	B	C
Positive	3 shared, 2 unshared	2 shared, 3 unshared	all 7 unshared
Neutral	all 5 unshared	all 5 unshared	2 shared, 1 unshared
Negative	all 5 unshared	4 unshared, 1 shared	all 3 shared
Total	16	16	17

Figure 1. Number of shared and unshared positive, neutral, and negative statements per Dean Candidate Profile (A, B, or C).

The assignment was designed in such a way, that from the perspective of the (simulated) Leader, candidate B was the most favorable option. However, from the perspective of the Follower, candidate B would be least favorable. For the Follower, candidate A was the best option, so Leader and Follower had conflicting initial preferences. To make matters more complicated, the unshared information would reveal that from the team perspective (if they decided to share their information and find out their information was not identical), candidate C was the best option (see figure 2. for a view of the distribution of attributes per candidate). Thus, for the Follower to arrive at the team's optimal decision,

(s)he would need to override (a) her/his own initial preference and (b) the Leaders' initial preference.

	Participant (Follower)			Confederate (Leader)			Combined (team)		
Candidate -->	A	B	C	A	B	C	A	B	C
Positive	5	2	3	3	5	4	5	5	7
Neutral	2	3	3	3	2	2	5	6	3
Negative	2	4	3	3	2	3	6	5	3
Total	9	9	9	9	9	9	16	16	17

Figure 2. Composition of Dean Candidate Profiles

Assessment of Statements for Composition of the Dean profiles (A, B, and C).

Prior to the experiment, a questionnaire was administered among business school students with the aim of collecting statements to construct the information sharing task that would be used to measure voice. The organizational scenario and information sharing game used in the experiment was going to be staged around the search for a new faculty dean. Followers would have to choose the best dean out of three candidates (A, B, or C), based on these candidates' statements. In order to create a task that would be important and relevant for students and thus create ecological validity, we surveyed students on how they would evaluate the statements of the fictional Dean candidates.

To develop the statements that would be used in this questionnaire, we first interviewed 10 students to evaluate what they perceived to be important subjects. From these interviews, we extracted 20 statements on what was of importance for a new dean.

Thereafter, we evaluated these 20 statements in a 45-minute interview session with the actual

Dean of our economics and business faculty. The dean evaluated the statements on (a) their relative importance to someone in a dean-position and (b) their practical feasibility. Examples of such statements are: “There should be more lockers available for students” (which was labelled as relatively unimportant) and “The deadlines for test grading should be more strictly maintained” (which was labelled as highly important and feasible). During the interview with the dean, a number of other questions were asked, such as: “If you would summarize the ideal dean attributes in five separate statements, what would they be?” All of these statements, including other more practical statements retrieved from the interview about for example normal Dean working hours, were counterbalanced (if possible), and incorporated in the final questionnaire.

Two-hundred and eighty three students (46% women, 54% men, average age = 20.28, $SD=1.88$, all years of enrolment) completed a 67 statement questionnaire through an online platform (Qualtrics.com). They were asked to evaluate these statements from their own (students’) point of view and rated the items on a 5-point scale going from negative (1) to positive (5). The scales were accompanied by an emoticon (an icon resembling a face), which changed expressions upon the answers to the statements. The mean ($M = 3.41$, $SD = 0.98$) of all statements was used as a reference point for neutral statements. All statements below 1 standard deviation below the mean (< 2.43) were marked as negative statements, all statements above 1 standard deviation above the mean (> 4.39) were marked as positive. Each dean candidate profile was composed of a combination of positive, neutral (all statements closest to the mean), and negative statements. These statements were balanced in terms of shared/unshared information across the Leader and Follower information (see Figure 1), and in terms of a content fit with each overall Dean profile (Dean A had a Scientific focus, Dean B had an Educational Focus, Dean C equally incorporated Science and Education).

Each statement printed in bold made it to the final profile selection (Appendix Tables A1 - A4).

Procedure, Measures and Tasks

Measures

Independent Variable WMC. After entering the experiment room, visual WMC was measured for each Participant separately in a quiet room on a laptop with the mark 'FOLLOWER'. All participants received an explanation of the test on screen, followed by four practice trials. WMC was assessed by means of a delayed serial recognition task which has been used in previous research to measure visual WMC (3,4,5). Participants performed a series of 96 trials. On each trial participants were presented with eight pictures (visual WMC) appearing sequentially in the center of the laptop screen. Each stimulus remained on screen for 250 milliseconds (ms), extending the complete trial over a period of 2000 ms. After a series of eight stimuli, the screen went blank for 1000 ms during which participants had to keep all information in memory (the average amount of visual objects people usually recall after 1000 ms. delay is 6.3 (3,4).

Following this “maintenance interval” they were shown a single picture. As a response to this last target stimulus, they had to indicate whether they had seen it in the previous series of eight stimuli, or not. In other words: is the target stimulus (1) a new stimulus, which *has not* appeared in the previous series, or is it (2) an old stimulus, which *has* appeared in the previous series. Fifty percent of the 8-word series were followed by an old target stimulus (no-change trials). The other fifty percent of the series was followed by a new target stimulus, which was different from the previous eight trials (change-trials). These new target stimuli were randomly drawn from a set of 130 possible items. Change versus no-

change trials were presented randomly. For each set we assessed response accuracy with a perfect WMC score of 96.

Working Memory Capacity was calculated with a linear transformation of accuracy into capacity as is used by Sligte and colleagues (3,4). WMC was calculated by transforming the percentage of correct responses into a capacity estimate between 0 and 8, correcting for chance level (the percentage of correct responses if participants randomly select a response, which is 0.5 or 50%). We used the following formula:

$$\text{WMC} = (\% \text{ correct responses} - \text{chance level}) * (n \text{ items in task} * 1 / 1 - \text{chance level})$$

This results in: $\text{WMC} = (\% \text{ correct} - 0.5) * (8 * 0.5)$. Performing at chance level would thus result in a capacity estimate of 0, and perfect performance would result in a capacity of 8.

The number of items participants could retain in working memory ranged from 2.33 to 6.83 ($M = 5.31$, $SD = .89$).

Organizational Setting. Since our experimental setting was staged around employee-manager voice, the organizational tasks needed to be completed in a team with a Follower (participant) and a team manager (Leader). Followers were first introduced to their team Leader after they completed their WMC task. The Leader would be (randomly chosen) one of two confederates who were both research associates at the university. Both confederates had teaching experience in small and large lectures, supervised theses, and were used to interaction with students in the role of expert, leader and mentor. To be able to control for gender differences, one confederate was male, the other was female. Both were blind to the study's hypotheses and did not know the participants prior to the experiment. After the introduction, the experimenter explained that the Leader (confederate) and the Follower (participant) would be doing some assignments together, that the pay-off in the end depended

upon the team's accomplishments, and that the leader was responsible for the team choices, and of distributing payment. Following, both received and signed an informed consent form.

Following this introduction, the team worked on a filler task together to strengthen their bond and create a climate for idea generation. Their task was to try to come up with possible things to do with a brick, with an emphasis on the idea being creative, but yet usable. An example of a creative and useful idea is: 'hollow the brick and use it as a vase'. Leaders received strict instructions to always react positively to an idea suggested by a Follower, regardless of their evaluation of the quality of the idea.

Dependent Variables: Measures of Voice and Voice Quality. After the Follower had read the instructions and had 5 minutes to read through his or her information, the Leader made contact through a computer chat-window using the following line: "Hello, this is not an easy assignment. I have tackled it as follows: I gave every attribute an evaluation (positive, neutral, negative). The experimenter is on his way with my notes. Based on this, I select candidate B: that one has most positive attributes, least neutral and least negative. The other two are comparable. So I choose B, you can look at my notes if you want to know why." The Leader did not specifically request a response, or ask for suggestions or ideas. The Follower was now in the possession of an information sheet of the Leader, with all attributes marked with either a green + (positive), a red -- (negative) or a blue N (neutral). Furthermore, the Followers had colored markers available if (s)he wished to organize his/her thoughts in a similar manner. So the Followers had full opportunity to find out that B was actually the least optimal choice, that A was only optimal from his/her own perspective, and that the best option for the team would be C.

We assessed voice *behavior* by giving the Follower two choices: 'Ok, I am fine with B' (no voice), or 'I think there is another possibility, let's talk' (voice). Please note that, knowing that the Leader would decide upon the pay-off, opposing the Leaders' opinion

directly by voicing, entailed social risk: voicing could mean that the Follower would lose salary. If the Follower chose the second option, the simulated Leader asked which would be the Followers preferred candidate. The Follower also got the possibility to explain why (s)he would choose differently, but this was not requested. *Voice quality* was consequently assessed by *what* Followers voiced. Candidate A was lower in quality because this merely reflected the Followers' initial preference and a sub-optimal choice considering all information. Switching to candidate C was considered a high-quality suggestion because this candidate was objectively the best team choice.

Our most important expectation was that because people high in WMC have the ability to focus their attention for long periods of time, they would be more able to overcome the tendency to go for their initial choice, incorporate more possibilities, oversee more information, be more persistent and therefore have higher chances to get to the best solution for the team. In other words: WMC was expected to be highest in the group that suggested candidate C to the Leader.

Checks

Confederates and Control variables. As mentioned earlier, confederates were either male or female. Participants (randomly) assigned to either of these Leaders showed no differences in control variables. Both Leaders were (equally) perceived as competent, intelligent, trustworthy, appreciative Leader figures who took charge of the task. Furthermore, no between-Leader differences were found on the two dependent variables (voice quantity or quality) or the independent variable WMC. Table 1 shows an overview of how Leaders were perceived by participants on a 1 to 7 Likert scale.

	Male Leader		Female Leader		Total	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
The leader seemed very capable of being a leader	5.56	0.91	5.79	0.96	5.67	0.93
My leader was taking charge of the task	5.00	1.04	5.44	1.11	5.21	1.09
To me, the leader seems to be an intelligent person	5.92	0.94	6.00	0.95	5.96	0.94
I trusted the leader	5.19	1.26	5.53	1.24	5.36	1.25
The leader created a safe environment for me to take initiative	5.28	1.06	5.59	1.21	5.43	1.14

Table 1. Evaluations of Leader characteristics by the participants on a 1 – 7 Likert scale.

APPENDIX STUDY 1

# statement	M	SD	Candidate A		Candidate B		Candidate C	
			Participant (Follower)	Confederate (leader)	Participant (Follower)	Confederate (leader)	Participant (Follower)	Confederate (leader)
Positive	1	4.71	0.46					1
	2	4.53	0.60			1		
	3	4.48	0.84	1	1			
	4	4.47	0.63	1				1
	5	4.45	0.50			1	1	
	6	4.43	0.84		1	1	1	
	7	4.42	0.66	1	1	1	1	
	8	4.41	0.59					1
	9	4.38	0.79			1		
	10	4.37	0.66	1	1			
	11	4.33	0.96					
	12	4.29	0.73	1				1

Table A1. Pilot evaluations of Positive statements and distribution across Dean Profiles.
Note. Only bold statements were used in the profiles.

# statement	M	SD	Candidate A		Candidate B		Candidate C	
			Participant (Follower)	Confederate (leader)	Participant (Follower)	Confederate (leader)	Participant (Follower)	Confederate (leader)
33	3.63	1.16						
34	3.63	1.08	1		1			
35	3.62	1.27						
36	3.58	1.08		1				
37	3.54	1.09						
Neutral 38	3.40	1.17						
39	3.39	1.17		1		1	1	1
40	3.37	1.00			1		1	1
41	3.26	1.43		1				
42	3.16	1.33				1	1	
43	3.04	1.22	1		1			

Table A2. Pilot evaluations of Neutral statements and distribution across Dean Profiles.
Note. Only bold statements were used in the profiles.

# statement	M	SD	Candidate A		Candidate B		Candidate C	
			Participant (Follower)	Confederate (leader)	Participant (Follower)	Confederate (leader)	Participant (Follower)	Confederate (leader)
50	2.49	1.12						
51	2.48	1.30						
52	2.46	1.16						
53	2.41	1.27					1	1
54	2.35	1.15						
55	2.34	1.46						
56	2.33	1.12						
57	2.32	1.11	1				1	1
58	2.23	1.01		1			1	1
59	2.16	0.96		1	1			
60	2.14	1.17	1		1			
61	2.10	0.91			1			
62	2.08	1.17						
63	2.04	1.02			1	1		
64	1.93	1.05		1				
65	1.68	0.73						
66	1.51	0.61				1		
67	1.25	0.73						

Table A3. Pilot evaluations of Negative statements and distribution across Dean Profiles.
Note. Only bold statements were used in the profiles.

TABLE A4. All statements used in the pilot survey and their average evaluations.

	#	M	SD	Statement text (in Dutch)
Pos	1	4.71	0.46	“De faculteit Economie en Bedrijfskunde moet (nog) beter worden: ik ga voor kwaliteit van onderwijs”
	2	4.53	0.60	“De ideale decaan communiceert goed”
	3	4.48	0.84	“Docenten zouden de deadlines van het teruggeven van tentamens strakker moeten naleven”
	4	4.47	0.63	“Een decaan hoort te communiceren met werkgevers binnen zijn/haar vakgebied om te zorgen dat kennis en vraag naar kennis in de toekomst op elkaar aansluiten”
	5	4.45	0.50	“Een ideale decaan kan goed omgaan met een veranderlijke omgeving als een universiteit”
	6	4.43	0.84	“Een goede decaan heeft een goed team om zich heen nodig, alleen kom je nergens”
	7	4.42	0.66	“Studentenevaluaties zijn belangrijk voor een faculteit”
	8	4.41	0.59	“Een decaan heeft als taak om zowel onderwijs als onderzoek optimaal te laten presteren”
	9	4.38	0.79	“Het is goed dat sommige colleges op meerdere momenten in de week gegeven worden: zo zijn ze gemakkelijker in te plannen voor werkende studenten”
	10	4.37	0.66	“Studenten moeten (nog) beter voorbereid worden op de arbeidsmarkt”
	11	4.33	0.96	“Er moet meer aandacht komen voor stage-programma’s. Afgestudeerden komen nu moeilijk aan een baan”
	12	4.29	0.73	“Het is belangrijk dat een decaan een academische achtergrond heeft; anders heeft hij/zij geen feeling met studenten en docenten”
	13	4.27	0.69	“Mijn taak is om te zorgen dat we gezamenlijk de faculteit weer opbouwen”
	14	4.26	0.73	“De ideale decaan kan strategisch denken”
	15	4.24	0.95	“Er moeten meer zelf-studie plekken komen.”
	16	4.22	0.88	“Studentenraden en Medezeggenschapsraden zijn belangrijk om naar te luisteren”
	17	4.21	0.78	“De ideale decaan is integer”
	18	4.17	0.68	“Wat onderwijs betreft veranderen we nu dingen maar we zijn er nog lang niet”
	19	4.16	0.96	“Er moeten meer ruimtes komen met computers om samen te werken; deze zijn nu bijna altijd vol”
	20	4.16	0.90	“Ik ben bereid om veel meer dan 38 uur te werken, dit hoort nu eenmaal bij deze verantwoordelijke functie”
	21	4.14	0.83	“Een decaan hoort ook oog te hebben van hoe kennis breder in de maatschappij inzetbaar is”
	22	4.12	0.71	“Vakken moeten (nog) beter op elkaar worden afgesteld, verschillende afdelingen kunnen beter samenwerken”
	23	4.10	0.83	“De ideale decaan is een goede onderhandelaar in de zin van het zoeken van overeenstemming tussen groepen”
	24	4.09	0.96	“Het zou goed zijn de wachttijd bij de informatie balie verminderen: deze is vaak wel een half uur”
	25	4.08	0.63	“Een Economie en Bedrijfskunde decaan hoort ook feeling te hebben met het bedrijfsleven”

	26	4.05	0.94	“Er moet veel aandacht besteed worden aan het werven van top-academici”
	27	4.04	1.09	“Kluisjes zijn niet de eerste prioriteit van een decaan”
	28	4.02	0.86	“De faculteit Economie en Bedrijfskunde moet (nog) beter worden: ik ga voor kwaliteit van onderzoek”
	29	3.98	0.87	“De ideale decaan is een team player”
	30	3.98	0.75	“Een ideale decaan kan goed omgaan met slimme mensen”
	31	3.96	1.05	“De roosters kunnen beter. Nu zijn er vaak in de propedeuse gaten van 3 tot 6 uur”
	32	3.92	1.13	“Colleges zouden vaker op dezelfde locatie in Amsterdam gegeven moeten worden”
Neu	33	3.63	1.16	“Colleges zouden minder in de avond moeten plaatsvinden”
	34	3.63	1.08	“Studenten moeten meer ambitie hebben voor buitenlandstages”
	35	3.62	1.27	“De hoorcolleges in de propedeuse zijn vaak zo massaal, er is behoefte aan meer afwisseling met kleine groepen”
	36	3.58	1.08	“Gaten in de roosters zouden door de studenten gebuikt moeten worden om te studeren en opdrachten te maken”
	37	3.54	1.09	“De ideale decaan is competitief”
	38	3.40	1.17	“Een goede decaan is afkomstig uit het bedrijfsleven”
	39	3.39	1.17	“Colleges worden nu eenmaal op verschillende locaties gegeven, dat heeft te maken met capaciteit”
	40	3.37	1.00	“Een decaan hoort zich te bemoeien met alle interne operaties”
	41	3.26	1.43	“Ik ben voor ‘selectie aan de poort’ voor de faculteit Economie en Bedrijfskunde”
	42	3.16	1.33	“Of er al dan niet met artikelen of een boek gewerkt wordt, is aan de desbetreffende docent”
	43	3.04	1.22	“De samenwerking tussen afdelingen is prima: onderwijsprogramma’s zijn goed op elkaar afgesteld”
	44	2.89	1.41	“Of een decaan een academische achtergrond heeft doet er niet toe: een goede leider is een goede leider”
	45	2.86	1.44	“Ik ben tegen ‘selectie aan de poort’ op de faculteit Economie en Bedrijfskunde” (CB)
	46	2.82	1.40	“Er zouden meer kluisjes beschikbaar moeten komen voor studenten”
	47	2.81	1.24	“Er gaan genoeg studenten naar het buitenland voor stages”
	48	2.77	1.31	“In de propedeuse moet meer met artikelen gewerkt worden”
	49	2.66	1.24	“Er zijn genoeg zelfstudie plekken op dit moment”
Neg	50	2.49	1.12	“De stage-programma’s zijn goed zoals ze zijn, studenten hebben genoeg ingangen in het bedrijfsleven”
	51	2.48	1.30	“Communicatie met werkgevers over consistentie tussen kennis en vraag naar kennis is geen prioriteit”
	52	2.46	1.16	“De ideale decaan trekt zijn eigen plan”

53	2.41	1.27	“De deadlines voor het teruggeven van tentamens worden nu goed genoeg nageleefd”
54	2.35	1.15	“De ideale decaan zoekt geen compromissen”
55	2.34	1.46	“Studentenevaluaties zijn er vooral voor de show”
56	2.33	1.12	“Kleine werkgroepen in de propedeuse zijn te duur”
57	2.32	1.11	“De wachttijd bij de informatiebalie heeft voor mij geen prioriteit”
58	2.23	1.01	“Colleges moeten gewoon eenmalig gegeven worden: studenten moeten hier omheen plannen”
59	2.16	0.96	“Studenten en Medezeggenschapsraden hebben wel een mening, maar een decaan moet zich hier niet door laten leiden.
60	2.14	1.17	“Samenwerken op computers kunnen studenten beter thuis doen dan op de faculteit”
61	2.10	0.91	“Studenten worden momenteel prima voorbereid op de arbeidsmarkt”
62	2.08	1.17	“Ik ben niet bereid om meer dan 38 uur te werken”
63	2.04	1.02	“Kwaliteit van onderzoek vind ik geen prioriteit voor onze faculteit”
64	1.93	1.05	“Feeling met het bedrijfsleven is als decaan niet belangrijk: een decaan heeft te maken met wetenschappers
65	1.68	0.73	“De link tussen de wetenschap en de maatschappij is voor een decaan niet belangrijk”
66	1.51	0.61	“Onze faculteit kan ook wel zonder top-academici”
67	1.25	0.73	“Kwaliteit van onderwijs vind ik geen prioriteit voor onze faculteit”
Totals	3.41	0.98	

Study 2

Method

Before explaining the details of this experiment, we ask you to envision a typical meeting room, with about 10 people in it. One of those people is your team leader, who is sitting at the head of the table. The leader is talking about how to change the organization, and you have some ideas of your own you would like to suggest. Other people are voicing their ideas as well, drawing your attention away from how to formulate your own ideas. Some of the ideas of others are actually good, and you see your leader's appreciation for these ideas. Other ideas are quite repetitive and do not seem to improve the discussion. While you listen to the ideas of others, you try to focus on which idea will still add novel and useful information to the discussion. What mentally happens while people try to make this decision, and how distraction of attention (thus WMC load) might affect such decisions, is what this second experiment is about. In the design of this experiment, we aimed to create a situation which matches the challenges described in the scenario above as closely as possible.

The main purpose of this experiment, was to investigate what mentally happens while people select ideas to voice in a social context. Specifically, we aimed to explore whether loading WMC with another task (thus distracting attention away from voice), would affect the number of voiced ideas (quantity), and the selection of original and useful ideas (voice quality). Furthermore, we wanted to study whether the possible effect of WMC (load) on quantity or quality of voice would also appear within individuals, as opposed to only between individuals (Study 1). We thus designed two within- person conditions in this experiment, one with, and one without load on working memory. Participants were presented with the opportunity to voice ideas, while executing another task at the same time. This other task was either extremely easy (no WMC load) or quite difficult (high WMC load). Voice quantity was reflected in the number of ideas participants voiced in each condition. Voice quality was

assessed **after** the experiment: all voice was rated by three independent raters in terms of whether or not voice provided novel and useful solutions superior to the status quo.

Procedure, Manipulations, Tasks, and Measures

Idea Generation Phase. The experiment started with an anonymous idea generation task from which participants could later select ideas to voice. For ecological validity, these should be ideas that mattered to the participants personally, so we asked them to anonymously generate 10 problems, opinions, or suggestions for the improvement of their faculty, a measure of creativity that is often used in experimental research (6). Participants had 10 minutes to generate their ideas. We told them explicitly that these ideas would not be evaluated or used for analysis, but that this was just an anonymous practice round. During idea generation, the experimenter digitally classified each idea (low/high quality) immediately after each idea was entered by the participant. The computer used these classifications for random-weighted message distribution across the two WMC Load conditions so both conditions contained an equal amount of high vs. low quality ideas. Later in the experiment, during the actual WMC load manipulation, participants saw a selection of 5 of those 10 ideas while being under No WMC load, and 5 of their own 10 ideas while being under High WMC load. The computer ensured that selection of ideas for a condition was random, with the restriction that each condition would have an equal distribution of low quality and high-quality ideas.

Dual tasks. During the following experimental phase, participants could engage in two types of payment dependent behaviors. There were two main types of tasks. One was the WMC load task, the other was the option to voice. Participants were requested to put their primary focus on the WMC load task, and this was also the task that paid a stable salary. Like in many organizations, participants thus had a core-task. The task that pays their basic salary.

Their core task, the working memory load task (counting backwards), always resulted in a €2.50 salary if performed correctly.

Secondary to this task, participants had the (extra-role) opportunity to voice, which could result in €1,- bonus (for high quality) or €1,- loss (for low quality) for every voiced suggestion, problem, or opinion depending on the leader's evaluation of voice quality. The leader role was played by the experimenter who sat in the same room as the participant, and who scored each suggestion as high / low quality on a computer. The experimenter was trained prior to the experiment by judging the full dataset of a prior study with the same task. The experimenter was first trained to judge both originality and usefulness separately, and in a second round of judgement on the same ideas, to make an intuitive judgement of both originality and usefulness in one quality judgement: high or low. These quality scores were used for 3 purposes: (1) to create a social evaluative context where the participant feels evaluated by the leader, (2) to make sure that the computer would be able to distribute an equal amount of good and bad ideas across the WMC load conditions, and (3) to make sure we could pay participants based on the instructions we gave them: €1,- bonus (for high quality) or €1,- loss (for low quality) for every voiced suggestion. These financial rewards or punishments are there to ensure that the content of voice matters and is consequential. As experimenter and participant were not bonded like real leaders and followers in organizations, we needed a financial incentive to make sure voice had consequences.

The experiment started with an exercise of the working memory-load task (counting backwards from 100, in steps of 2) followed by performance feedback. We explained that this task would be repeated several times during the experiment, and that this would always be their primary task with a standard reward of €2.50. Next, participants practiced to perform the WMC load task while having to select text from a menu of words (easy) and sentences (harder). We designed two tasks in which participants needed to select items from a pool of

10 words ('select the 3 largest animals from this list') or 10 sentences ('select the 5 true statements about animals'). We started with the simple task, followed by the more complex task. While they executed these tasks, they performed a working memory load task (counting backwards). The purpose of this practice phase was to familiarize participants with the combination of counting while selecting messages, limiting possibilities for technical confusion during the real experimental phase.

Dependent Variables: Communication (voice) phase and simultaneous load manipulation. Following the practice phase, participants received the instruction that they would get to see a random selection of their own ideas, as well as a random selection of other people's ideas. We added 5 ideas from a database of relevant ideas to each Load condition because voice takes place in a social context. Quite often, voice happens while being subjected to the ideas and opinions of other people, and thus we decided to take a selection of ideas on the same task that were generated in a previous experiment. We made sure each quality score (1 to 6) was represented once in each condition.

These other-ideas were kept constant between participants: each Load condition contained the full quality (low – high) range of ideas as coded by two independent judges. During the 'window for voice', participants thus saw 10 messages from which they could select those ideas they found 'worthy' to voice: 5 of their own, and 5 generated by others. We told them that they would get the opportunity to voice any idea to the experiment-leader, who would evaluate it and pay them accordingly. Voice (as opposed to the generation phase) was thus no longer anonymous and moreover, financially risky: voicing a good idea would result in a €1,- gain, voicing a bad idea in a €1,- loss. Participants were not pushed to voice any ideas, because their primary and most profitable task was to count backwards, either from 80 in steps of 2 (High Load) or from 100 in steps of 0, thus repeating the same number vocally (Low Load). Participants performed two trials with 10 messages each and a random

order of High vs. Low Load. Thus, participants had the opportunity to voice their previously generated ideas, *while* performing a WMC-load task. Voicing ideas was done digitally, by selecting and sending the message to the experimenter, who would calculate payment upon evaluation of the idea at the end of the experiment accordingly.

Dependent variables voice quality and quantity. *Voice quantity* was measured by counting the number of voiced ideas in each Load condition. Specifically, we counted how many suggestions, problems, or opinions participants were sent to the experimenter (quantity). Across conditions, participants voiced 2.87 ideas on average (*range* = 0 - 10, *SD* = 2.29). Further, whereas Study 1 measured voice quality as an objective operationalization of value and insight, we now used expert ratings of voice quality. After the experiment, all ideas from the generation phase were evaluated by three independent judges, who rated whether the ideas provided novel solutions or useful implementations superior to the status quo, on a scale from 0 (not at all) to 6 (highly) as is prescribed in creativity research (6). Inter-rater reliability was reasonable for the generated quality (*ICC* = .659). Voice quality was the average quality of those ideas that were voiced by the participants, which was higher ($M_{\text{voiced}} = 3.06$, $se = .036$, $\text{within}CI_{95} = [2.99, 3.13]$) than the average quality of the ideas they generated prior to the voice phase ($M_{\text{generated}} = 2.86$, $se = .036$, $\text{within}CI_{95} = [2.78, 2.93]$), $F(1, 64) = 7.76$, $p = .007$, $\eta^2 = .108$. Within-participant confidence intervals were calculated using the method developed by Loftus and Mason, as revised by Cousineau (7).

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